

ePGD Linear Regression Tutorial

Problem Solving Session

Problem 1: Understanding the Regression Setup

Consider the following dataset:

Student	Study Hours	Attendance (%)	Exam Score
1	2	60	50
2	3	70	58
3	5	75	68
4	6	85	78
5	8	90	88

Suppose we want to predict Exam Score using Study Hours and Attendance.

Answer the following:

- Write the output vector Y .
- Identify the predictor variables.
- Find the number of observations n .
- Find the number of predictors p .
- Construct the design matrix X including an intercept column.
- State the dimensions of X and Y .

Hint

The response variable forms Y . The predictors form the columns of X . Include a column of ones for the intercept.

Direct Answers

a)

$$Y = \begin{bmatrix} 50 \\ 58 \\ 68 \\ 78 \\ 88 \end{bmatrix}$$

b)

$$x_1 = \text{Study Hours}, \quad x_2 = \text{Attendance}$$

c)

$$n = 5$$

d)

$$p = 2$$

e)

$$X = \begin{bmatrix} 1 & 2 & 60 \\ 1 & 3 & 70 \\ 1 & 5 & 75 \\ 1 & 6 & 85 \\ 1 & 8 & 90 \end{bmatrix}$$

f)

$$X : 5 \times 3, \quad Y : 5 \times 1$$

Problem 2: Simple Linear Regression

A fitted regression model for predicting examination score from study hours is

$$\hat{y} = 35 + 6x,$$

where x is the number of hours studied.

Calculate the predicted examination score for

$$x = 2, \quad x = 4, \quad x = 6, \quad x = 8.$$

Also answer:

- What does the intercept 35 represent?
- What does the slope 6 represent?

Hint

Substitute each value of x into

$$\hat{y} = 35 + 6x.$$

Direct Answers

x	$\hat{y}$
2	47
4	59
6	71
8	83

a)

$$\beta_0 = 35 : \text{ predicted score when } x = 0$$

b)

$$\beta_1 = 6 : \text{ predicted score increases by 6 per extra study hour}$$

Problem 3: Residuals and RSS

Suppose the actual outputs are

$$Y = \begin{bmatrix} 10 \\ 15 \\ 20 \\ 25 \end{bmatrix}$$

and the predicted outputs are

$$\hat{Y} = \begin{bmatrix} 11 \\ 14 \\ 19 \\ 27 \end{bmatrix}.$$

Calculate:

- a) The residual vector

$$e = Y - \hat{Y}.$$

- b) The squared residual for each observation.

- c) The Residual Sum of Squares

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Hint

Use

$$e_i = y_i - \hat{y}_i.$$

Then square the residuals and add them.

Direct Answers

- a)

$$e = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -2 \end{bmatrix}$$

- b)

$$[1, 1, 1, 4]$$

- c)

$$RSS = 7$$

Problem 4: Closed-Form OLS Solution

Consider

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

and

$$Y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

Using

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

calculate the following step by step:

- a) X^T
- b) $X^T X$
- c) $(X^T X)^{-1}$
- d) $X^T Y$
- e) $\hat{\beta}$

Hence determine

$$\hat{\beta}_0 \quad \text{and} \quad \hat{\beta}_1$$

and write the fitted regression equation.

Hint

Calculate in the order

$$X^T \rightarrow X^T X \rightarrow (X^T X)^{-1} \rightarrow X^T Y \rightarrow \hat{\beta}.$$

Direct Answers

a)

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

b)

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

c)

$$(X^T X)^{-1} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix}$$

d)

$$X^T Y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}$$

e)

$$\hat{\beta} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}$$

$$\hat{\beta}_0 = \frac{1}{3}, \quad \hat{\beta}_1 = \frac{3}{2}$$

$$\hat{y} = \frac{1}{3} + \frac{3}{2}x$$

Problem 5: Matrix Dimensions

Suppose

$$X \in \mathbb{R}^{100 \times 4}$$

and

$$Y \in \mathbb{R}^{100 \times 1}.$$

Determine the dimensions of:

$$X^T,$$

$$X^T X,$$

$$(X^T X)^{-1},$$

$$X^T Y,$$

and

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

Also determine how many regression coefficients are being estimated.

Hint

Use

$$(a \times b)(b \times c) = a \times c.$$

Direct Answers

$$\begin{aligned} X^T &: 4 \times 100, \\ X^T X &: 4 \times 4, \\ (X^T X)^{-1} &: 4 \times 4, \\ X^T Y &: 4 \times 1, \\ \hat{\beta} &: 4 \times 1 \end{aligned}$$

$$\text{Number of coefficients} = 4$$

Problem 6: Full Column Rank

Consider

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 6 \\ 1 & 4 & 8 \end{bmatrix}.$$

Notice that the third column satisfies

$$X_3 = 2X_2.$$

Answer:

- Are the columns of X linearly independent?
- What is the rank of X ?
- Does X have full column rank?
- Can $(X^T X)^{-1}$ exist?
- Explain why the ordinary closed-form OLS expression creates a problem in this case.

Hint

Check whether one column can be written as a multiple of another.

Direct Answers

a)

No

b)

$\text{rank}(X) = 2$

c)

X does not have full column rank

d)

No

e)

$X^T X$ is singular, so $(X^T X)^{-1}$ does not exist

Problem 7: Predictions from OLS Coefficients

Suppose an OLS calculation gives

$$\hat{\beta} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The design matrix is

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

Calculate

$$\hat{Y} = X\hat{\beta}.$$

Hence write the predicted value corresponding to each value of x .

Hint

Use

$$\hat{Y} = X\hat{\beta}.$$

Direct Answers

$$\hat{Y} = \begin{bmatrix} 5 \\ 8 \\ 11 \\ 14 \end{bmatrix}$$

x	$\hat{y}$
1	5
2	8
3	11
4	14

$$\hat{y} = 2 + 3x$$

Problem 8: Prediction and Residual Interpretation

Suppose

$$Y = [20, 25, 30, 35, 40]$$

and

$$\hat{Y} = [21, 24, 29, 36, 39].$$

Calculate the residual for every observation.

Then answer:

- a) Which observations have positive residuals?
- b) Which observations have negative residuals?
- c) What does a positive residual indicate?
- d) What does a negative residual indicate?

Hint

Calculate

$$e_i = y_i - \hat{y}_i.$$

Direct Answers

$$e = [-1, 1, 1, -1, 1]$$

a)

Observations 2, 3, 5

b)

Observations 1, 4

c)

Actual value > predicted value

d)

Actual value < predicted value

Problem 9: Hat Matrix

Consider

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

and

$$Y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}.$$

Calculate

$$H = X(X^T X)^{-1} X^T.$$

Then calculate

$$\hat{Y} = HY.$$

Also calculate $\hat{Y}$ using

$$\hat{Y} = X\hat{\beta}.$$

Verify that the two prediction vectors are identical.

Hint

Use

$$H = X(X^T X)^{-1} X^T$$

and

$$\hat{Y} = HY.$$

Direct Answers

$$H = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{2}{6} \end{bmatrix}$$

$$HY = \begin{bmatrix} \frac{13}{6} \\ \frac{11}{3} \\ \frac{31}{6} \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}, \quad X\hat{\beta} = \begin{bmatrix} \frac{13}{6} \\ \frac{11}{3} \\ \frac{31}{6} \end{bmatrix}$$

$$HY = X\hat{\beta}$$

Problem 10: TSS, MSS and RSS

Suppose an OLS regression gives

$$Y = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 7 \end{bmatrix}, \quad \hat{Y} = \begin{bmatrix} 2.1 \\ 3.7 \\ 5.3 \\ 6.9 \end{bmatrix}.$$

Calculate:

- a) The mean $\bar{y}$.
- b) The Total Sum of Squares

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2.$$

- c) The Model Sum of Squares

$$MSS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

- d) The Residual Sum of Squares

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

- e) Verify

$$TSS = MSS + RSS.$$

Hint

First calculate

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

Then apply the definitions of TSS, MSS and RSS.

Direct Answers

- a) $\bar{y} = 4.5$
- b) $TSS = 13$
- c) $MSS = 12.8$
- d) $RSS = 0.2$
- e) $TSS = MSS + RSS = 12.8 + 0.2 = 13$

Problem 11: Coefficient of Determination R^2

A regression model gives

$$RSS = 40$$

and

$$TSS = 200.$$

Calculate

$$R^2 = 1 - \frac{RSS}{TSS}.$$

Then answer:

- What percentage of the variation in Y is explained by the regression model?
- Which model gives a better fit:

$$R^2 = 0.80$$

or

$$R^2 = 0.45?$$

- Why should a high R^2 still be interpreted with caution when extra predictors are added?

Hint

Use

$$R^2 = 1 - \frac{RSS}{TSS}.$$

Direct Answers

$$R^2 = 0.80$$

a)

$$80\%$$

b)

$$R^2 = 0.80$$

c)

Adding predictors can increase R^2 even if they are not useful

Problem 12: Statistical Regression Model

Consider

$$y_i = 5 + 3x_i + \epsilon_i$$

with

$$\epsilon_i \sim N(0, 4).$$

Determine:

- a) β_0^*
- b) β_1^*
- c) σ^2
- d) σ

If $x_i = 4$ and the observed noise happens to be

$$\epsilon_i = -2,$$

calculate the observed value y_i .

Hint

Compare

$$y_i = 5 + 3x_i + \epsilon_i$$

with

$$y_i = \beta_0^* + \beta_1^* x_i + \epsilon_i.$$

Also,

$$N(0, \sigma^2) = N(0, 4).$$

Direct Answers

- a) $\boxed{\beta_0^* = 5}$
- b) $\boxed{\beta_1^* = 3}$
- c) $\boxed{\sigma^2 = 4}$
- d) $\boxed{\sigma = 2}$

For $x_i = 4$ and $\epsilon_i = -2$:

$$\boxed{y_i = 5 + 3(4) - 2 = 15}$$

Problem 13: OLS and Maximum Likelihood

Suppose the statistical model is

$$y_i = x_i^T \beta + \epsilon_i,$$

where

$$\epsilon_i \sim N(0, \sigma^2).$$

For two possible coefficient estimates, the corresponding RSS values are

$$RSS(\beta_A) = 25$$

and

$$RSS(\beta_B) = 40.$$

Under the Gaussian noise assumption:

- Which coefficient estimate would OLS prefer?
- Which coefficient estimate would Maximum Likelihood Estimation prefer?
- What does this tell you about the relationship between OLS and MLE under Gaussian noise?

Hint

OLS minimizes RSS. Under Gaussian additive noise, MLE also corresponds to minimizing RSS.

Direct Answers

a)

$$\boxed{\beta_A}$$

b)

$$\boxed{\beta_A}$$

c)

$$\boxed{\hat{\beta}_{OLS} = \hat{\beta}_{MLE}}$$

Problem 14: Estimating Noise Variance

A regression model contains

$$n = 30$$

observations and

$$p = 2$$

predictors.

Suppose

$$RSS = 108.$$

Calculate

$$\hat{\sigma}^2 = \frac{RSS}{n - (p + 1)}.$$

Then calculate

$$\hat{\sigma}.$$

Hint

Use

$$\hat{\sigma}^2 = \frac{RSS}{n - (p + 1)}.$$

Then take the square root.

Direct Answers

$$\hat{\sigma}^2 = \frac{108}{30 - (2 + 1)} = \frac{108}{27}$$

$$\hat{\sigma}^2 = 4$$

$$\hat{\sigma} = 2$$

Problem 15: Standard Error

Two studies estimate the same regression coefficient.

Study A gives

$$\hat{\beta}_1 = 2.5, \quad SE(\hat{\beta}_1) = 0.20.$$

Study B gives

$$\hat{\beta}_1 = 2.5, \quad SE(\hat{\beta}_1) = 0.80.$$

Answer:

- a) Which study gives the more precise estimate?
- b) Which study has greater uncertainty?
- c) Calculate the intervals

$$\hat{\beta}_1 \pm SE(\hat{\beta}_1)$$

for both studies.

Hint

A smaller standard error means a more precise estimate.

Direct Answers

a)

Study A

b)

Study B

c) Study A:

[2.30, 2.70]

Study B:

[1.70, 3.30]

Problem 16: p-value and Statistical Significance

Suppose a multiple regression produces the following results:

Predictor	Coefficient	p-value
TV	0.045	0.0001
Radio	0.180	0.002
Newspaper	0.003	0.82

Using

$$\alpha = 0.05,$$

determine:

- a) For which predictors should the null hypothesis

$$H_0 : \beta_j = 0$$

be rejected?

- b) For which predictor should the null hypothesis not be rejected?

- c) Which predictors show statistical evidence of association with Sales?

Hint

Compare each p-value with

$$\alpha = 0.05.$$

Use

$$p < 0.05 \Rightarrow \text{Reject } H_0.$$

Direct Answers

- a)

TV and Radio

- b)

Newspaper

- c)

TV and Radio

Predictor	Decision
<i>TV</i>	Reject H_0
<i>Radio</i>	Reject H_0
<i>Newspaper</i>	Fail to reject H_0

Problem 17: Simple Regression Case Study

Suppose the fitted model is

$$\widehat{Sales} = 7.03 + 0.0475(TV).$$

Calculate the predicted Sales when

$$TV = 100$$

and

$$TV = 200.$$

Then calculate the increase in predicted Sales when TV advertising increases from 100 to 200.

Interpret the coefficient

$$0.0475.$$

Hint

Substitute the TV value into

$$\widehat{Sales} = 7.03 + 0.0475(TV).$$

Direct Answers

For $TV = 100$,

$$\widehat{Sales} = 11.78$$

For $TV = 200$,

$$\widehat{Sales} = 16.53$$

Increase:

$$16.53 - 11.78 = 4.75$$

Coefficient:

$$0.0475 = \text{increase in predicted Sales per unit increase in TV}$$

Problem 18: Multiple Linear Regression

Suppose the fitted model is

$$\widehat{Sales} = 2.94 + 0.046(TV) + 0.189(Radio) - 0.001(Newspaper).$$

A company spends

$$TV = 150,$$

$$Radio = 20,$$

and

$$Newspaper = 30.$$

Calculate the predicted value of Sales.

Then answer:

- a) What does the coefficient 0.046 for TV mean?
- b) What does the coefficient 0.189 for Radio mean?
- c) What does the negative Newspaper coefficient indicate?

Hint

Substitute all three predictor values into the multiple regression equation.

Direct Answers

$$\widehat{Sales} = 2.94 + 0.046(150) + 0.189(20) - 0.001(30)$$

$$\boxed{\widehat{Sales} = 13.59}$$

a)

+0.046 Sales units per TV unit, holding Radio and Newspaper fixed

b)

+0.189 Sales units per Radio unit, holding TV and Newspaper fixed

c)

-0.001 Sales units per Newspaper unit, holding TV and Radio fixed

Problem 19: Correlation

Consider the following correlation matrix:

	TV	Radio	Newspaper	Sales
TV	1.00	0.05	0.06	0.78
Radio	0.05	1.00	0.35	0.58
Newspaper	0.06	0.35	1.00	0.23
Sales	0.78	0.58	0.23	1.00

Answer:

- Which predictor has the strongest correlation with Sales?
- Which predictor has the weakest correlation with Sales?
- Rank TV, Radio and Newspaper from strongest to weakest correlation with Sales.
- Which two advertising predictors have the strongest correlation with each other?

Hint

Compare the magnitudes of the relevant correlation coefficients.

Direct Answers

a)

TV ($r = 0.78$)

b)

Newspaper ($r = 0.23$)

c)

TV > Radio > Newspaper

d)

Radio and Newspaper ($r = 0.35$)

Problem 20: Maximum Likelihood Estimation with Laplace Noise

Consider the following variation of the statistical model for linear regression. We assume that the i th training output y_i is related to the i th training input x_i via

$$y_i = \langle \beta^*, x_i \rangle + \epsilon_i,$$

where ϵ_i represents independent additive noise described by the probability density function

$$f_\epsilon(t) = \frac{1}{2}e^{-|t|}.$$

Naturally, β^* is unknown and needs to be estimated from the n training data samples.

Answer the following:

- a) Write down the likelihood function

$$\mathcal{L}(\beta)$$

associated with the training data.

- b) Express the Maximum Likelihood Estimate of β^* as the solution of an optimization problem.
c) Is this estimator identical to the ordinary least-squares solution?

Hint

Use

$$\epsilon_i = y_i - \langle \beta, x_i \rangle$$

inside

$$f_\epsilon(t) = \frac{1}{2}e^{-|t|}.$$

For independent observations, multiply the individual densities.

Direct Answers

- a)

$$\mathcal{L}(\beta) = \left(\frac{1}{2} \right)^n \exp \left[- \sum_{i=1}^n |y_i - \langle \beta, x_i \rangle| \right]$$

- b)

$$\hat{\beta}_{MLE} = \arg \min_{\beta} \sum_{i=1}^n |y_i - \langle \beta, x_i \rangle|$$

c)

$$\hat{\beta}_{MLE} \neq \hat{\beta}_{OLS} \quad \text{in general}$$

$$\text{Laplace noise} \Rightarrow L_1, \quad \text{Gaussian noise} \Rightarrow L_2$$

Problem 21: Salary Regression with a Categorical Predictor

We have the following data on employee salaries in an organization.

Employee ID	Years of Experience	Department	Salary
1	1	Sales	\$52,500
2	3	Sales	\$57,500
3	5	Sales	\$62,500
4	8	Sales	\$70,000
5	1	Tech	\$62,500
6	3	Tech	\$67,500
7	5	Tech	\$72,500
8	8	Tech	\$80,000

Your task is to build a linear regression model to predict Salary using:

- Years of Experience
- Department

Answer the following:

- Describe the regression model mathematically.
Clearly define the dummy variable used to represent Department and interpret each fitted parameter.
- Using your fitted model, predict the salary of an employee in the **Sales** department with

2

years of experience.

Hint

Take Sales as the reference category:

$$D = \begin{cases} 0, & \text{Sales} \\ 1, & \text{Tech.} \end{cases}$$

Use

$$\text{Salary} = \beta_0 + \beta_1 \text{Experience} + \beta_2 D.$$

Direct Answers

a)

$$\widehat{\text{Salary}} = 50000 + 2500(\text{Experience}) + 10000D$$

where

$$D = \begin{cases} 0, & \text{Sales} \\ 1, & \text{Tech.} \end{cases}$$

$$\beta_0 = 50000$$

$$\beta_1 = 2500$$

$$\beta_2 = 10000$$

β_0 : predicted Sales salary at 0 years experience

β_1 : \$2500 increase per additional year of experience

β_2 : Tech salary is \$10,000 higher than Sales at equal experience

b) For Sales,

$$D = 0.$$

Therefore,

$$\widehat{Salary} = 50000 + 2500(2) + 10000(0)$$

$$\widehat{Salary} = \$55,000$$